Deep Learning methods for Radar Target Detection: A Comparative Study of detection methods on Raw and Range-Doppler Data

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***Abstract*—Cell-averaging constant false alarm rate (CA-CFAR) detection methods remains the baseline for radar target detection, yet its performance degrades sharply at low signal-to-noise ratio (SNR) and its threshold estimation is sequential manner. WE propose and evaluate six convolutional neural network (CNN) detectors against CA-CFAR baseline on simulated pulse-Doppler radar data cubes, spanning two input representations – raw in-phase/quadrature (I/Q) samples and range-Doppler (RD) magnitude maps – at three depths (3, 4 and 6 convolutional layers), each one 5000-sample training set over -10 to 13 dB SNR and identical per-SNR test sets. The representations diverge sharply. RD-CNNs reach 90% probability of detection (P_d) with less SNR as compared to CA-CFAR, with no further benefits. raw I/Q CNNs never exceed ≈0% P_d at any depth or SNR despite carrying 5–10× more parameters. RD networks overshoot the design false-alarm rate ($P_{fa} = 10^{-5}$), reaching a frame-level P_{fa} of ≈7.7% at -10 dB before converging above -2 dB, an effect traced to the fixed CNN threshold's sensitivity to absolute noise power. CNN inference takes 0.6–2.7 ms per frame against 91.9 ms for serial CA-CFAR, a 34–150× speedup. RD-CNNs thus offer a modest detection gain and a large latency gain once their false-alarm calibration is corrected, while the raw I/Q representation as trained here did not yield a functional detector.**

***Index Terms*—Radar target detection, convolutional neural networks, CA-CFAR, range-Doppler processing, probability of detection, false alarm rate, pulse-Doppler radar.**

I. Introduction

Constant false alarm rate (CFAR) detection has been the workhorse of radar signal processing for four decades. Cell-averaging CFAR (CA-CFAR) estimates a local noise level from reference cells around the cell under test (CUT) and sets an adaptive threshold maintaining a design false-alarm probability under homogeneous noise. It is simple and analytically well understood, but degrades at low SNR, in non-homogeneous clutter, and in multi-target scenes where reference-window contamination biases the noise estimate. Deep learning instead learns noise and target statistics directly from data without a parametric model, and convolutional neural networks (CNNs) have shown promise for radar detection, classification and range-Doppler (RD) map interpretation – motivating a systematic comparison against the classical baseline under matched conditions.

Classical CFAR variants – cell-averaging (CA), ordered-statistic (OS) and greatest-of/smallest-of CFAR – trade sensitivity, false-alarm regulation and clutter-edge robustness through the reference-cell combining rule. They are cheap and analytically tractable but assume locally stationary noise and process one RD cell at a time, limiting low-SNR sensitivity and throughput [1]–[3]. Chen et al. [4] survey the shift to learned detection, noting that the input representation remains an open design question.

Most learned detectors replace the CFAR threshold with a CNN on a pre-processed RD map. Lee et al. [5] classified targets from RD maps, establishing them as a natural image-like input, as did Patel et al. [6] for automotive spectra. RadCNN [10] consumes RD ambiguity-function patches in place of a 2-D CFAR sweep at improved low-SNR detection and lower cost; DL-CFAR [11] learns the decision from the reference-window statistics CA-CFAR would combine analytically; CFARNet [12] folds a false-alarm constraint into the training objective, retaining constant false-alarm behaviour at improved P_d; Liang et al. [15] adapted a lightweight YOLOv4-tiny backbone for RD images; and Jiang et al. [9] posed detection as CNN classification on processed data. Throughout, the RD map is the default input, always studied in isolation. A smaller literature asks whether a CNN can detect from raw samples instead. Gu et al. [14] applied a CNN to raw I/Q samples for pulse detection, outperforming a short-time Fourier transform detector and several learned baselines, showing that range and Doppler structure can be recovered implicitly – but never comparing against an RD-map CNN of matched capacity. Chen et al. [13] separately proposed a target-attentional CNN down-weighting non-target range bins, though such designs target *recognition* rather than detection versus CFAR.

CNN detectors are almost always evaluated on a single input representation; most studies evaluate one architecture, leaving depth unexamined; raw I/Q and RD inputs are rarely compared at matched capacity; and none are compared at matched depth against the *same* CA-CFAR baseline on identical per-SNR data. It is therefore unclear how much of the reported CNN

advantage comes from the representation rather than capacity.

This paper presents (i) a reproducible framework generating matched raw I/Q and RD representations of a single-target-in-noise scenario across a wide SNR sweep; (ii) trains six CNNs – three depths per representation – and evaluates them against a CA-CFAR baseline on identical test data at every SNR point, comparing P_d, P_{fa} and latency; and (iii) quantifies the latency advantage of GPU-batched inference over serial CA-CFAR. Sections II–V cover the system model, detection framework, results and conclusions.

II. System Model

Consider a data cube spans N_s range (fast-time) samples, N_p pulses (slow-time) and N_c receive channels (Table I). Rather than assume range compression applied, the signal is simulated at raw ADC level with an explicit linear-frequency-modulated (LFM) chirp, compressed afterwards by matched filtering (Section II). The reference chirp of length L, sampled at f_s with bandwidth B and duration $T_p = L/f_s$, is

$$c[n] = \exp\left(j\pi K \left(\tfrac{n}{f_s}\right)^2\right),\ n = 0, \ldots, L-1,\ K = \tfrac{B}{T_p}. \quad (1)$$

Noise is drawn independently per channel, pulse and range sample as circularly symmetric white Gaussian noise,

$$n[m,k] \sim \tfrac{1}{\sqrt{2}}\left(\mathcal{N}(0,\sigma^2) + j\,\mathcal{N}(0,\sigma^2)\right), \quad (2)$$

with m indexing pulses and k range bins. The variance σ^2 is set from a specified *post-compression* SNR by backing off the matched-filter gain of the L-sample chirp,

$$\mathrm{SNR}_{\mathrm{raw}} = \mathrm{SNR} - 10\log_{10} L, \qquad \sigma^2 = 10^{-\mathrm{SNR}_{\mathrm{raw}}/10}, \quad (3)$$

so that after matched filtering the range-compressed target-to-noise ratio is approximately the requested SNR. This excludes the additional $\approx 10\log_{10} N_p$ Doppler-FFT gain; the RD-CNN and CA-CFAR share the same post-FFT map and benefit equally, so the sweep is a fair common basis. The raw I/Q branch receives no compression gain, so its effective per-sample SNR is $\mathrm{SNR}_{\mathrm{raw}}$, about 19 dB below the nominal value for $L = 80$.

With a target present, a delayed, Doppler-modulated unit-amplitude copy of the chirp is added coherently across all N_c channels at random bins r, v:

$$s[m,k] = c[k-r]\exp\left(j2\pi\tfrac{(v-N_p/2)m}{N_p}\right),\ k \in [r, r+L), \quad (4)$$

and $s[m,k] = 0$ elsewhere; the $-N_p/2$ offset centres the Doppler index on the zero-Doppler bin, matching the `fftshift` convention of Section II. The received cube is $x[m,k] = n[m,k] + s[m,k]$, or $n[m,k]$ alone for pure-noise frames. Placement bounds keep targets clear of the cube edges; critically, $r_{\max}$ also guarantees the full L-sample echo fits inside the fast-time window, since truncated echoes would receive less energy than central ones regardless of SNR.

For each pulse m the fast-time samples are correlated against a Hamming-windowed reference chirp $w[n]c[n]$ by FFT-based correlation zero-padded to $N_{\mathrm{fft}} = 2^{\lceil \log_2(N_s+L-1)\rceil}$, keeping the first N_s outputs as the compressed profile $y[m,k]$. A Doppler FFT across m (with `fftshift`) then concentrates a moving target into a compact RD cell, giving the path echo → range compression → Doppler FFT → RD map → detector. Fig. 2 shows this on one frame.

TABLE I
Key Radar Simulation Parameters

Parameter	Symbol	Value
Range (fast-time) samples	N_s	256
Pulses (slow-time)	N_p	32
Receive antenna channels	N_c	3
Sampling frequency	f_s	20 MHz
Chirp length / bandwidth	L, B	80 samples, 10 MHz
Chirp duration	T_p	4 μs
Per-frame dimensionality	$N_pN_sN_c$	24,576 (complex)
Target range bin	r	$\in [8, 168)$
Target Doppler bin	v	$\in [4, 28]$
SNR sweep	SNR	-10–13 dB, 1 dB steps
Training set size	N_{train}	5000 (balanced)
Per-SNR test set (mixed)	$N_{\mathrm{test}}^{\mathrm{mix}}$	400 frames
Per-SNR test set (noise)	$N_{\mathrm{test}}^{\mathrm{noise}}$	400 frames
Calibration set (pure noise)	N_{calib}	10,000 frames

The raw I/Q branch taps the chain *before* both operations, receiving $x[m,k]$ with real and imaginary parts stacked across channels, so it must learn *both* the chirp compression and the Doppler structure implicitly – a strictly harder problem, and the reason the branches are not two views of an equally processed signal.

Two representations are produced per cube: a raw I/Q tensor of shape $2N_c \times N_p \times N_s$ stacking real and imaginary parts of $x[m,k]$, and an RD magnitude tensor of shape $N_c \times N_p \times N_s$ from the chain of Section II applied per channel. The N_c channels share a coherent return but independent noise, so the antenna dimension gives noise averaging, not angular information. A frame's nominal SNR is sampled independently of target presence, so target-absent frames receive the same noise scaling and the training set spans the sweep in both classes. Training uses N_{train} balanced examples with SNR drawn uniformly in 1 dB steps; independent per-SNR test sets of $N_{\mathrm{test}}^{\mathrm{mix}}$ mixed and $N_{\mathrm{test}}^{\mathrm{noise}}$ pure-noise frames are generated at each point with no reuse of placements or noise realizations. A held-out calibration set of N_{calib} pure-noise frames, whose noise level spans the full sweep, is generated once and reused at every SNR point to set all thresholds (Section IV-B).

Because each per-SNR noise test set uses one fixed noise level while the calibration set mixes levels, the two noise-floor distributions differ even though one threshold applies to both. This is scale-invariant for CA-CFAR, whose threshold is a ratio to the local reference mean, but not for the CNNs, and Section IV-D shows it is the mechanism behind the RD-CNN's elevated low-SNR false-alarm rate. Fig. 1 summarizes the three paths and the shared protocol.

III. Proposed Method

Detection is posed as binary classification over a whole radar frame, with six CNNs spanning two input domains

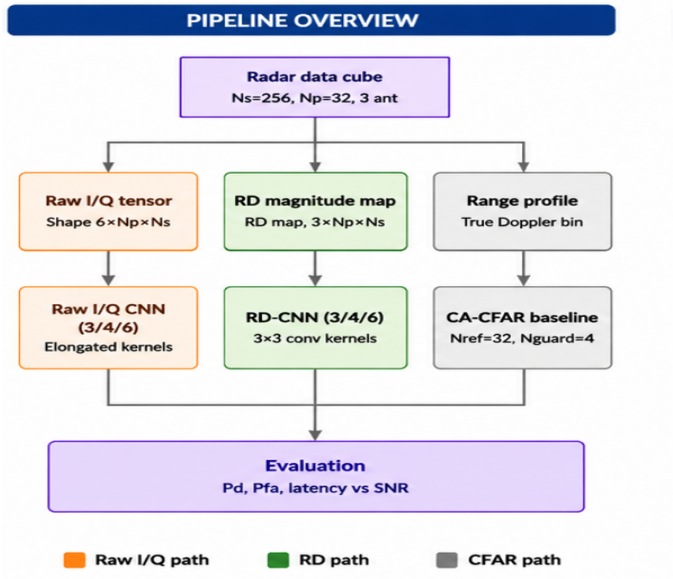


Fig. 1. The three detection paths. One radar cube becomes a raw I/Q tensor, an RD magnitude map and a range profile at the true Doppler bin, feeding the Raw I/Q CNN, RD-CNN and CA-CFAR detectors, all calibrated on a shared noise set and evaluated on the same metrics.

and three depths, trained and evaluated under the protocol of Section IV.

A. Raw I/Q CNN

The raw I/Q networks (RawIQCNN-3/4/6) consume the 6-channel real/imaginary tensor directly (Fig. 3) using rectangular kernels elongated along the fast-time axis, each followed by batch normalization, ReLU and strided downsampling. The geometry follows from Eq. (4): the target signature spans an $L = 80$-sample chirp beginning at the unknown bin r. An elongated kernel such as 5×15 in Conv 1 gives a receptive field wide enough to capture part of the chirp's frequency sweep, where a small isotropic kernel would see only samples indistinguishable from noise; deeper layers assemble this into an approximate matched-filter response. The asymmetric stride $S(1, 2)$ preserves full slow-time resolution, where only $N_p = 32$ pulses resolve the Doppler phase, while downsampling the longer fast-time axis. Depth is added with stride-1 refinement layers and channels widened to 256; global average pooling and a classifier ($256 \rightarrow 128 \rightarrow 2$, 50% dropout) give the logits.

B. Range-Doppler CNN

The RD networks (RD-CNN-3/4/6) consume the 3-channel RD magnitude map with 3×3 convolutions and max-pooling, halving spatial resolution while doubling channels from 32 to 128 (Fig. 4); global max pooling and a two-layer classifier ($128 \rightarrow 64 \rightarrow 2$, 40% dropout) produce the logits. Small isotropic kernels suffice because the RD transform already concentrates target energy into a compact spot.

C. Architectural Specifications

Per-depth geometries appear in Tables II and III. The raw I/Q branch uses global average pooling (GAP) and the RD branch global max pooling (GMP) instead of flattening: GAP aggregates the diffuse evidence of the pre-compression tensor, GMP isolates the compact peak of the RD transform. Both also make the representation invariant to *where* the response appears, since r and v are random and a flattened map would force a separate boundary per target position – leaving only *whether* a strong response exists, which is the frame-level decision trained.

At matched depth the raw I/Q network carries 5–10× more parameters than its RD counterpart, the wider channels offsetting the elongated kernels. This matters for Section IV-D: the raw I/Q failure is not a capacity shortfall, since the smallest raw I/Q network (1.02 M) already outsizes the largest RD network (0.30 M), pointing to a representational or optimization bottleneck.

TABLE II
RAW I/Q CONVOLUTIONAL NEURAL NETWORK ARCHITECTURES

Layer	Conv-3	Conv-4	Conv-6
Conv 1	5×15, 64, S(1,2)	5×15, 64, S(1,2)	5×15, 64, S(1,2)
Conv 2	3×11, 128, S(2,2)	3×11, 128, S(1,1)	3×11, 64, S(1,1)
Conv 3	3×7, 256, S(2,2)	3×7, 256, S(2,2)	3×7, 128, S(2,2)
Conv 4	–	3×5, 256, S(1,1)	3×5, 128, S(1,1)
Conv 5	–	–	3×3, 256, S(2,2)
Conv 6	–	–	3×3, 256, S(1,1)
Pool	Global adaptive average pooling $(1, 1)$		
Dense 1	Linear → 128 units, ReLU, dropout 0.5		
Dense 2	Linear → 2 units (classifier)		
Trainable params	1.02 M	2.01 M	1.50 M

Entries read kernel, output channels, stride S(x,y). Every convolution is followed by BatchNorm2d and ReLU.

TABLE III
RANGE–DOPPLER CONVOLUTIONAL NEURAL NETWORK ARCHITECTURES

Layer	Conv-3	Conv-4	Conv-6
Conv 1	3×3, 32, MP(2)	3×3, 32, MP(2)	3×3, 32, –
Conv 2	3×3, 64, MP(2)	3×3, 64, MP(2)	3×3, 32, MP(2)
Conv 3	3×3, 128, MP(2)	3×3, 128, MP(2)	3×3, 64, –
Conv 4	–	3×3, 128, MP(2)	3×3, 64, MP(2)
Conv 5	–	–	3×3, 128, –
Conv 6	–	–	3×3, 128, MP(2)
Pool	Global adaptive max pooling (1×1)		
Dense 1	Linear → 64 units, ReLU, dropout 0.4		
Dense 2	Linear → 2 units (classifier)		
Trainable params	0.10 M	0.25 M	0.30 M

MP(2) denotes a 2×2 MaxPool2d layer; "–" indicates no pooling. Every convolution is followed by BatchNorm2d and ReLU.

D. Training Procedure

All six models are trained independently in PyTorch [7] for E epochs with Adam [8] (learning rate η, weight decay λ), a schedule halving η every E_{step} epochs, batch size B and cross-entropy loss, on the same training set and shuffling policy (Table IV). A model is retrained up to three times from a fresh initialization if its training-set class-score separation (mean target minus mean no-target score) falls below 0.15; the first attempt clearing the threshold is kept. Measured on training scores, this safeguard did not ensure the raw I/Q boundary generalized to held-out data.

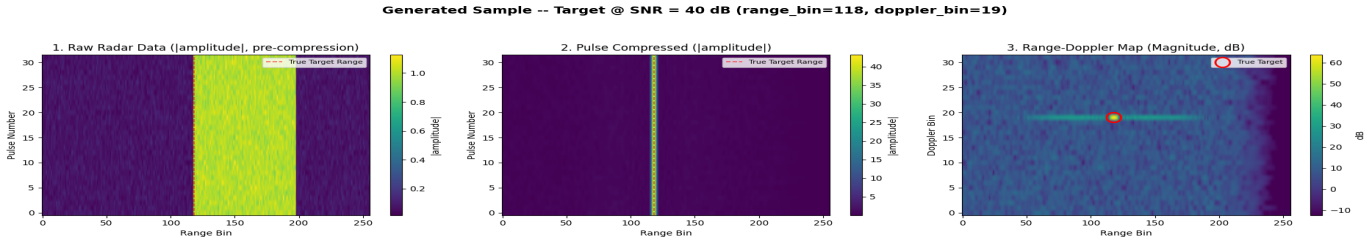


Fig. 2. Simulated single-antenna frame at SNR $= 40$ dB, target at $(r, v) = (118, 19)$. **(a)** Raw magnitude before compression: the $L = 80$-sample chirp echo spans range samples 118–197 in every pulse. **(b)** Pulse-compressed magnitude, energy concentrated at the target range bin. **(c)** RD magnitude map, the target an isolated peak. Dashed lines and the circle mark the true location.

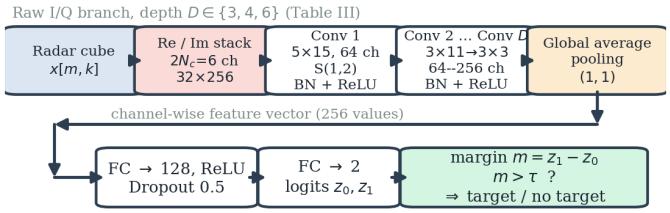


Fig. 3. Raw I/Q CNN pipeline. The 6-channel real/imaginary tensor passes through elongated-kernel blocks (Table II) before global average pooling and a two-layer classifier.

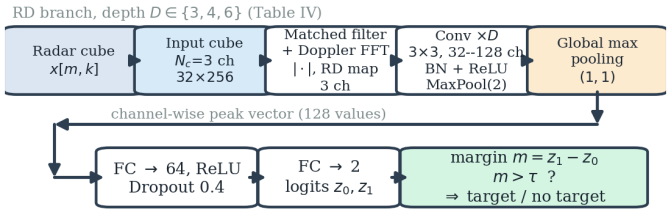


Fig. 4. RD-CNN pipeline. The RD magnitude map from the front end of Section II passes through isotropic 3×3 blocks (Table III) before global max pooling and a two-layer classifier.

IV. Performance Analysis

A. Baseline Detector (CA-CFAR)

The baseline operates on the range profile at the true Doppler bin with N_{ref} reference and N_{guard} guard cells per side of the CUT (Fig. 5). Its adaptive threshold is

$$T = \alpha \cdot \frac{\mu_{\text{left}} + \mu_{\text{right}}}{2}, \quad \alpha_0 = 2N_{ref}\left(P_{fa}^{-1/2N_{ref}} - 1\right), \quad (5)$$

where μ_{left}, μ_{right} are the mean powers of the two reference windows and P_{fa} is the design false-alarm probability. The closed form α_0 assumes i.i.d. exponential reference cells; since the empirical noise deviates from this ideal, α_0 serves only as an initial guess and is refined by bisection on the calibration set until the empirical cell-level false-alarm rate matches P_{fa}. Detection is evaluated over M Monte Carlo trials per SNR point: P_d at the true target cell, P_{fa} by sweeping every range bin at every Doppler bin of pure-noise frames (Table IV).

B. Threshold Calibration

All seven detectors are calibrated to the same P_{fa} on the shared N_{calib} pure-noise frames, independent of training and test data. Calibration runs once, and each threshold is held fixed at every SNR point – the CNN thresholds are *not* re-adapted per SNR. For CA-CFAR, α is calibrated at cell level

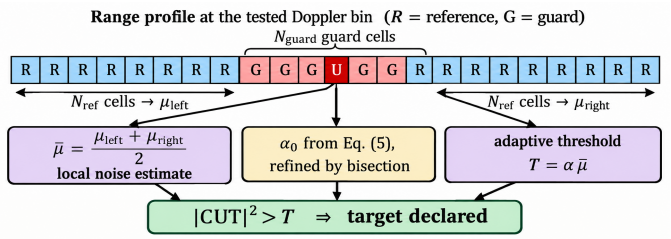


Fig. 5. CA-CFAR pipeline. Mean power from the two reference windows sets the adaptive threshold T of Eq. (5), against which the CUT is compared.

TABLE IV
Training Hyperparameters and CA-CFAR Configuration

Parameter	Symbol	Value
Training (all six CNNs)		
Training epochs	E	30
Initial learning rate	η	10^{-3}
Weight decay	λ	10^{-4}
LR-halving interval	E_{step}	10 epochs
Batch size	B	32
Optimizer / loss	–	Adam / cross-entropy
CA-CFAR baseline		
Reference cells (per side)	N_{ref}	32
Guard cells (per side)	N_{guard}	4
Design false-alarm prob.	P_{fa}	10^{-5}
Monte Carlo trials per SNR	M	100

over all $N_p N_s N_{\text{calib}}$ reference-window cells by bisection. For each CNN, the margin $m = z_1 - z_0$ (target minus no-target logit) is computed on the calibration set and the threshold set as

$$\tau = \bar{m} + z_{P_{fa}}\,\hat{\sigma}_m, \qquad z_{P_{fa}} = \Phi^{-1}(1 - P_{fa}), \qquad (6)$$

with $\bar{m}$, $\hat{\sigma}_m$ the empirical mean and standard deviation of the margin and Φ^{-1} the standard normal quantile function; a frame is declared a detection if its margin exceeds τ.

The families are not equally verifiable. CA-CFAR is calibrated over tens of millions of cells, so its empirical P_{fa} can be checked against 10^{-5} with hundreds of expected false alarms, whereas each CNN yields one frame-level decision per sample, giving only 0.1 expected false alarms on $N_{\text{calib}} = 10{,}000$ frames – not verifiable at this size. Eq. (6) is retained as the more defensible option at this scale.

C. Simulation Settings and Metrics

At each SNR point the CNNs see an identical mixed test set of $N_{\text{test}}^{\text{mix}}$ frames and a pure-noise set of $N_{\text{test}}^{\text{noise}}$ frames, using the fixed thresholds of Section IV-B. P_d is the fraction

of target-present frames whose CNN margin exceeds τ, or whose CUT exceeds T for CA-CFAR. P_{fa} comes from the pure-noise set but *not* on the same basis for both families: cell-level for CA-CFAR (fraction of swept cells firing) and frame-level for the CNNs; the two share axes for comparison against the common target but are not statistically equivalent, as one CNN decision aggregates all cells of a frame. *Latency* is the per-frame wall-clock time over 100 frames after GPU warm-up, with CUDA synchronization after each pass, GPU-batched for the CNNs and serial per-Doppler-bin for CA-CFAR.

D. Results and Discussion

Fig. compares each depth against CA-CFAR; Fig. aggregates depths per domain and adds the latency benchmark.

Probability of Detection : CA-CFAR rises from ≈0% at -10 dB through ≈65% at -4 dB to ≈89% at -3 dB, crossing 90% near -3 dB. The RD-CNN follows a similar S-curve shifted lower – ≈15–20% at -10 dB, ≈76–79% at -5 dB, ≈96% at -4 dB – crossing 90% near -4 dB, a gain of roughly 1 dB. This is real but modest, far smaller than the 8–10 dB sometimes suggested by less tightly controlled comparisons: under a strict $P_{fa} = 10^{-5}$ target and matched test data the advantage narrows. The raw I/Q CNN stays at ≈0% P_d across the *entire* sweep at every depth.

Depth has little effect either way: the three RD depths track each other with no consistent ordering, so the RD representation is already discriminative enough that extra layers do not improve P_d, while raw I/Q is depth-insensitive for the opposite reason, all three depths remaining near 0%.

False Alarm Rate: CA-CFAR tracks its design rate across the sweep by construction, appearing flat at the bottom of the plotted range. RD-CNN P_{fa} overshoots badly at low SNR, reaching ≈4.5–7.7% at -10 dB – three to four orders of magnitude above target – then decaying to ≈0% by roughly -2 dB. This is not measurement noise but a consequence of the calibration/test noise-floor mismatch of Section II: since lower nominal SNR means *higher* absolute noise power (Eq. 3), a fixed threshold calibrated against mixed noise levels fires most often where noise power is greatest. CA-CFAR is immune because its threshold is a *ratio* to the local reference mean and hence scale-invariant; the CNN margin is not. Raw I/Q P_{fa} stays near 0% throughout, but given its ≈0% P_d on the same data this reflects a degenerate classifier assigning nearly every input to the no-target class, not calibration.

It is therefore the RD-CNN, not the raw I/Q CNN, that shows the large false-alarm violation, and the mechanism is the fixed threshold rather than the representation. Detection gains thus cannot be assessed apart from false-alarm calibration: a scale-normalized or per-SNR-recalibrated threshold would be needed before deployment.

Computational Complexity : Fig. IV-D(e) gives per-frame inference times of 1.3, 2.7 and 1.1 ms for Raw IQ Conv-3/4/6 and 0.6, 0.7 and 0.9 ms for RD Conv-3/4/6, against 91.9 ms for serial CA-CFAR – a 34–150× speedup. RD-CNNs are consistently faster at matched depth, consistent with their smaller parameter counts. Two caveats apply: the CNN timings benefit from GPU batching while the reference CFAR loops over all 32 Doppler bins in Python, so a vectorized CFAR would narrow the gap; and the CNN latency excludes an equivalent multi-Doppler sweep. Even so, CNN detection – particularly the RD branch, both faster and the only functional detector here – is viable in real-time pipelines.

TABLE V
DETECTION SENSITIVITY AND FALSE-ALARM CALIBRATION BY FAMILY

Detector	**SNR @ 90% P_d**	**P_{fa} vs. 10^{-5} target**
CA-CFAR	≈ -3 dB	Matches target (by construction)
RD-CNN (3/4/6)	≈ -4 dB	Up to ∼4 orders of magnitude above target at -10 dB; converges above -2 dB
Raw IQ-CNN (3/4/6)	Not reached (P_d ≈0%)	≈0%, a degenerate always-"no-target" classifier, not calibration

Limitations. The scenario is a single target in white Gaussian noise, without clutter, multipath or multiple targets, flattering both families. CNN P_{fa} is frame-level and CA-CFAR P_{fa} cell-level, so the shared axes are not a matched-basis measurement; per-cell normalization is left to future work. The CNN threshold is calibrated once and held fixed rather than adapted per SNR, cannot be verified at the present calibration-set size, and is the source of the RD-CNN low-SNR excursion. The cause of the raw I/Q failure remains open – too few epochs or too little data, an unfavourable optimizer schedule, or a genuine representational ceiling – and separating these was outside this scope. Finally, the latency comparison pits a GPU-batched CNN against a single-core CFAR and is an upper bound on the speedup.

V. CONCLUSION

We demonstrated Six CNN detectors – two representations at three depths were compared against a CA-CFAR baseline on identical simulated data at every SNR point, all thresholds calibrated once to $P_{fa} = 10^{-5}$. The representations diverged sharply. RD-CNNs improved the SNR for 90% P_d by ≈1 dB and cut per-frame latency by 34–150×, but their false-alarm rate exceeded the design target at low SNR, an effect traced to the fixed threshold's sensitivity to absolute noise power. Raw I/Q CNNs, despite 5–10× more parameters at matched depth, never exceeded chance-level detection; whether the cause is optimization, data volume or a representational limit is left open. Since these results come from a single-target, clutter-free simulation with an SNR-independent threshold, the RD-CNN gains are an upper bound and the raw I/Q outcome a negative result specific to this configuration. Future work includes clutter-inclusive multi-target scenarios, Neyman–Pearson-calibrated thresholds, and matched-basis cell-level and hardware-matched latency comparisons.

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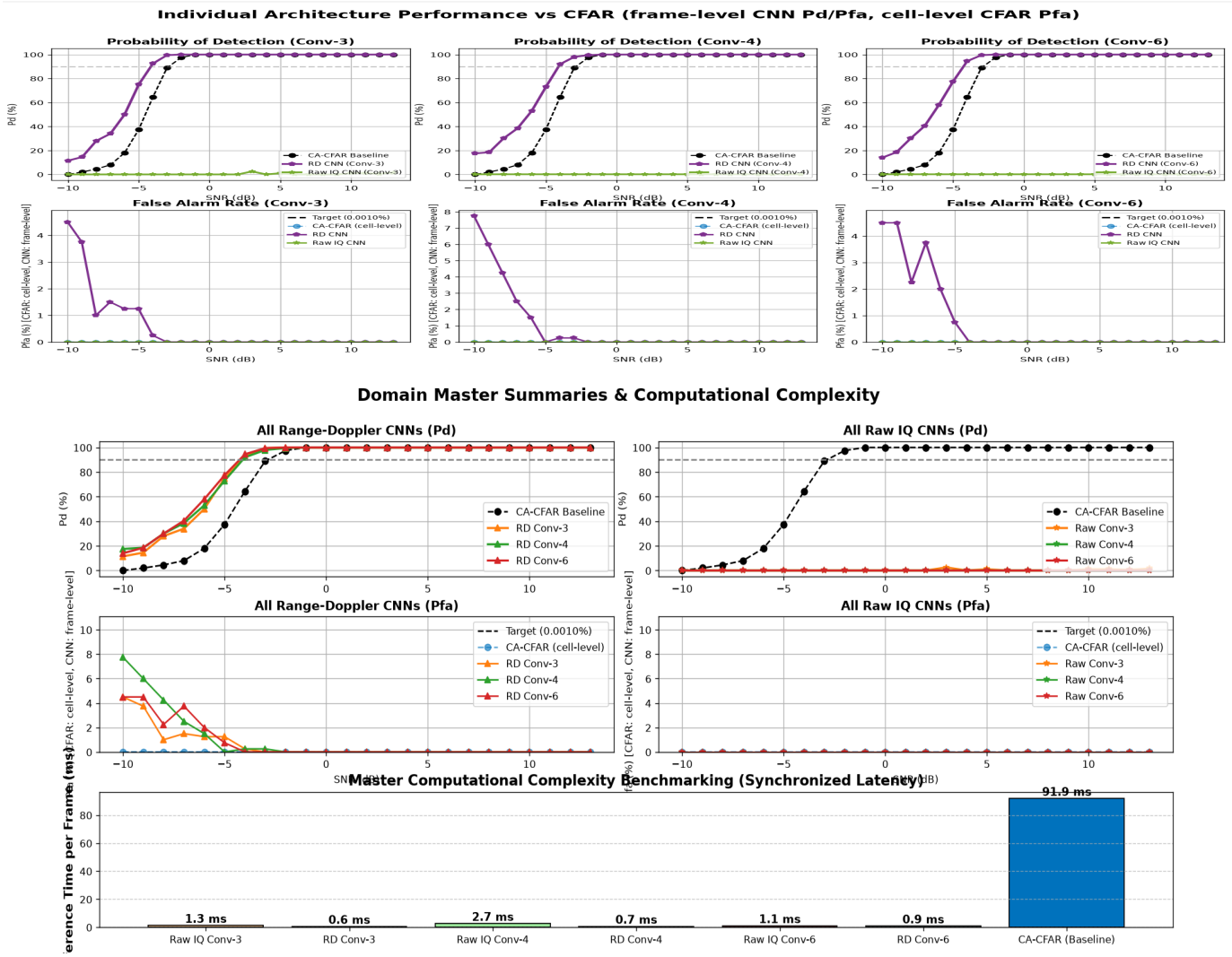


Fig. 6. Experimental results. Top: probability of detection and false-alarm rate versus SNR for Conv-3, Conv-4 and Conv-6 compared with CA-CFAR for both input representations. Bottom: aggregated P_d and P_{fa} by input domain and per-frame inference latency for all seven detectors.

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